

GITCG Universal Protocol V1.0

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1 Introduction

GITCG Universal Protocol, also known as *GIUP*, is a descriptive definition of game Genius Invokation TCG¹. GIUP is committed to creating a universal, transferable definition of situations and actions for GITCG. GIUP isn't a rulebook for GITCG - it's not trying to be the boss here. Instead, GIUP's goal is to describe the current state of play in GITCG and the potential action space, without getting tangled up in how different elements interact. So, we claim it *descriptive*.

2 Basic Elements

This section will introduce the key terms in GITCG and provide precise definitions.

Term 1: Player

GIUP uses a number starting from zero to represent players. In the basic version, there are only two options: 0 or 1. In the extended version, there will be other options, such as a random player (2).

Term 2: Current Player

Current Player represents the one who can take an action at now.

Term 3: Round

It is exactly same as Round in the game.

Term 4: Turn

When current player changes from 0 to 1 or 1 to 0, Turn +1. At the beginning of each Round, Turn is set to 1.

This definition already takes special cases into account.

Term 5: Element

Elements are:

0. Omni (also Empty for aura)
1. Pyro

2. Hydro
3. Electro
4. Cryo
5. Dendro
6. Anemo
7. Geo

Term 6: Damage Type

Damage Types not only include **Element** (Term 5), but also three additional options: piercing, physical and healing. Omni damage has no practical meaning here.

Healing is literally not a type of damage... but this definition makes life easier in many cases.

Term 7: HP

HP is a non-negative integer.

Term 8: Energy

Energy is a non-negative integer.

Term 9: Dice

Dice is a pair: (**Element** (Term 5) or Black or White, Number)

We use term **Elemental Dice** to refer dice with concrete **Element** (Term 5).

Term 10: Dice Pattern

Dice Pattern is an unordered list of **Dice** (Term 9).

An example is [*Pyro*, 1, *Black*, 2], which is a typical setting for Normal Attack of Pyro character.

Term 11: Skill

A skill consists of

1. Skill Name. An identifier(string).
2. **Energy** (Term 8).
3. **Dice Pattern** (Term 10).

¹https://en.wikipedia.org/wiki/Genius_Invokation_TCG

4. Type. Normal Attack, Elemental Skill or Elemental Burst.

Term 12: Usage

Usage is a non-negative integer or positive infinity. It usually indicates how many times some effect can be triggered. For a permanent effect, it's positive infinity.

Term 13: Superscript

Superscript is a **Usage** (Term 12) annotated in the upper right corner of some entity in game.

Term 14: Subscript

Subscript is a **Usage** (Term 12) annotated in the lower right corner of some entity in game.

Term 15: Zone

A Zone consists of:

1. Metadata. A collection of information that is not indexed by Route:
 - (a) Name
 - (b) Additional information specific to the Zone type
2. Entity List. An ordered list of components that can be indexed by Route. Each component is either:
 0. A Unit (any non-Zone entity)
 1. Another Zone

The ordering of Entity List is significant as it determines the Route indices. Unless specified otherwise, the ordering is fixed or follows FIFO for variable-length lists.

Term 16: Equipment

1. ID.
2. Description. Text.
3. Type. Weapon, Talent, Artifact, Technique or Other.

Term 17: Buff

1. ID.
2. Text description.
3. **Subscript** (Term 14).

Term 18: Character

A Character consists of:

1. Metadata:

- (a) ID
- (b) Aura (**Element** (Term 5)). Empty by default
- (c) **HP** (Term 7)
- (d) **Energy** (Term 8)

2. Entity List:

0. Equipment Zone (ordered by type: Weapon, Talent, Artifact, Technique, Other)
1. Skill Zone
2. Buff Zone

Term 19: Character Zone

A Character Zone consists of:

1. Metadata:

- (a) Player ID
- (b) Active character index

2. Entity List:

0. Buff Zone
1. Character List (ordered list of **Character** (Term 18))

Term 20: Support Zone

A Support Zone consists of:

1. Metadata:

- (a) Player ID
- (b) Maximum capacity (4)

2. Entity List:

0. Ordered list of Support entities, where each Support entity contains:
 - (a) Metadata:
 - i. Name
 - ii. **Subscript** (Term 14)
 - iii. **Superscript** (Term 13)

Term 21: Summon Zone

A Summon Zone consists of:

1. Metadata:

- (a) Player ID
- (b) Maximum capacity (4)

2. Entity List:

0. Ordered list of Summon entities, where each Summon entity contains:

- (a) Metadata:
 - i. Name
 - ii. **Damage Type** (Term 6)
 - iii. **Subscript** (Term 14)
 - iv. **Superscript** (Term 13)

Term 22: Dice Zone

A Dice Zone consists of:

1. Metadata:
 - (a) **Player** (Term 1).
 - (b) Count K.
2. Entity List:
 0. Ordered list of **Elemental Dice**

Term 23: Card

1. ID.
2. Text description.
3. **Energy** (Term 8).
4. **Dice Pattern** (Term 10).

Term 24: Hand Zone

A Hand Zone consists of:

1. Metadata:
 - (a) Count K.(10 maximum)
2. Entity List:
 0. Ordered list of **Card** (Term 23)

Term 25: Pile

A Pile consists of:

1. Metadata:
 - (a) **Player** (Term 1).
 - (b) Count K.
2. Entity List:
 0. Ordered list of **Card** (Term 23), where each card has:
 - (a) Position (-1 for unknown, or 0 to M-1 from top to bottom)

Term 26: Player Area

A Player Area consists of:

1. Metadata:
 - (a) **Player** (Term 1) ID
 - (b) Arcane Legend status
2. Entity List:
 0. Character Zone
 1. Support Zone
 2. Summon Zone
 3. Dice Zone
 4. Hand Zone
 5. Pile

Term 27: Game

A Game consists of:

1. Metadata:
 - (a) List of **Player** (Term 1)
2. Entity List:
 0. Two **Player Area** (Term 26)

Term 28: Route

Each unit/Zone is indexed by a unique route.

A Route is a series of indices separated by dots. It starts from **Game** (Term 27) and goes down until reaching the **Zone** (Term 15)/Unit you want to specify.

For example, 0.2 refers to the Summon Zone of Player 0.

0.2.1 refers to the 2nd summon of Player 0.

For cards in the pile, indices directly correspond to their actual order, but you typically shouldn't specify Cards in the pile directly by index unless you know their actual positions.

In implementation, we recommend binding the corresponding Route to all entities that have been described. This can also be referred to as their Address in the game.

3 Game Flow

The Game starts off with **Player** (Term 1) 0. **Current Player** (Term 2) can perform Actions. The rule of player switch is out of the scope of GIUP.

Term 29: Action

1. Type. Select, End, Confirm, Surrender.

2. Route. The target's route which is selected. None for other types of actions.

This definition is highly abstract because we declare GIUP to be descriptive. We will explain how it can be properly implemented to correctly execute normal GITCG operations. For example, when you want to play a card from your hand, you first need to use Select to choose that card, then repeatedly use Select to choose the corresponding dice, and finally use Confirm to execute. Since Route indexes all selectable entities, you can always complete the same operations as in GITCG through Select (just like clicking on your phone), and then finally execute it through Confirm. The definition itself doesn't involve these implementation details, so it is implementation-independent.

4 Random Extension

This section we introduce a random player and describe how to control the random events. This extension is denoted as GIUP-R.

Term 30: Random Player

We use -1 to refer to this random player.

After the execution of Confirm **Action** (Term 29), **Current Player** (Term 2) is changed to -1.

Since the rules for action execution are beyond the scope of GIUP, we need to additionally state here how to properly implement the random player to control random events. For instance, when Player 0 needs to draw cards, the Current Player will be switched to Random Player, and the Random Player decides which cards are drawn. This extension will be supplemented later to cover possible Random Player actions. The most significant benefit of doing this is that players can make predictions by controlling the random player - for example, players can calculate optimal strategies by assuming which cards the opponent holds (assume random player has an action: SetOppoHand).

Glossary

1	Player	1
2	Current Player	1
3	Round	1
4	Turn	1
5	Element	1
6	Damage Type	1
7	HP	1
8	Energy	1
9	Dice	1
10	Dice Pattern	1
11	Skill	1
12	Usage	2
13	Superscript	2
14	Subscript	2
15	Zone	2
16	Equipment	2
17	Buff	2
18	Character	2
19	Character Zone	2
20	Support Zone	2
21	Summon Zone	2
22	Dice Zone	3
23	Card	3
24	Hand Zone	3
25	Pile	3
26	Player Area	3
27	Game	3
28	Route	3
29	Action	3
30	Random Player	4